

PHOTOMETRY OF GALAXIES AND STAR CLUSTER ANALYSIS

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Abstract

This report explores two important concepts in observational astronomy - photometry and star cluster analysis. In the first part of the report, we have demonstrated the various parts of a data pipeline we have created to do aperture photometry. This includes data reduction, astrometry, finding robust apertures, and finally photometry. One of the main results in this report was finding the magnitude of a Seyfert 2 galaxy with identifier MCG 01-05-47, which came out to be 14.429 ± 1.172 . We have also found spectral energy distributions of some sources in this section.

The second part of this report focuses on the analysis of the M 46 open cluster. This section includes an analysis to determine the membership of stars, Hertzsprung-Russell diagrams, and an interpretation of the diagram.

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Chapter 1

Introduction

Astronomical photometry is a fundamental technique in modern astrophysics. It provides crucial insights into the intrinsic properties of stars and galaxies, enabling the determination of their temperatures, luminosities, distances, and chemical compositions. Complementing this, the study of star clusters offers a unique opportunity to understand stellar evolution, as their member stars are typically co-eval and co-distant, simplifying the analysis of their life cycles. This report delves into the methodologies of astronomical photometry and star cluster analysis, contributing to our understanding of stellar populations and the dynamic nature of the cosmos.

1.1 Astronomical Photometry

Astronomical Photometry refers to the measurement of flux or brightness of astronomical objects. Light captured from a telescope using a photometer - in our case, a CCD (Charged Coupled Device) Photometer converts light into an electric current by the photoelectric effect. Photometry is usually done by passing light through optical passband which allow for comparison of observations across wavelengths.

1.2 Aperture Photometry

The basic principle of aperture photometry is to sum up the observed flux within a given radius from the centre of an object, then subtract the total contribution of the sky background within the region, leaving only the flux from the object.

1.3 Apparent and Absolute Magnitude

Apparent magnitude is the luminosity of an object as observed from Earth. Absolute magnitude is equal to the apparent magnitude that the object would have when viewed from a distance of 10 parsecs from the Earth. Both apparent and absolute magnitudes are defined on a reverse logarithmic scale, meaning that brighter objects have a lower magnitude number, and the brightest objects have a negative magnitude. For example, the sun has a magnitude of -26.83.

1.4 Relative Photometry

Relative Photometry is a technique used to measure the apparent magnitude of an astronomical object by comparing it to one or more nearby sources with known magnitudes. This technique helps correct for instrument sensitivity and atmospheric extinction, and temporal variation that occurs due to far-away objects being compared.

1.5 Hertzsprung-Russell Diagrams

Hertzsprung-Russell or HR diagrams are scatter plots showing the relation between the magnitudes and effective temperatures of stars in a cluster. The diagram was independently developed by Ejnar Hertzsprung and Henry Norris Russell in 1911 and 1913, respectively. An HR diagram is instrumental in mapping stellar evolution, as the location occupied by a particular star in the diagram signifies its age and type.

1.6 Data Used

Data used from photometry was captured by the 1.3-m Devasthal Fast Optical Telescope (Joshi et al. [2022](#)) installed at Devasthal, Nainital and operated by the Aryabhata Research Institute of Observational Sciences (ARIES), an autonomous institute under the Department of Science and Technology (DST). The data for the Hertzsprung-Russell diagrams was sourced from the European Space Agency's Gaia mission.

Building upon these foundational concepts, this report outlines a comprehensive project encompassing both photometric analysis and star cluster characterisation. Chapter 2 details the development and application of a robust data reduction pipeline, crucial for preparing raw astronomical images for analysis. Chapter 3 then elaborates on the photometric techniques employed and presents results from their application to a Seyfert 2 galaxy. The construction and interpretation of Spectral Energy Distributions (SEDs) are discussed in Chapter 4. Chapter 5 focuses on the in-depth analysis of the M46 star

cluster, including the rigorous iterative filtering method used for member identification and the resulting Hertzsprung-Russell diagram. Finally, Chapter 6 summarises the key conclusions drawn from this work and outlines avenues for future research.

Chapter 2

Data Reduction for Photometry

In this project, we have created a pipeline for data reduction, including bias and flat correction, background subtraction, and then detection and cataloguing of sources in the image for easier reference. Before putting data through the pipeline, we first perform astrometry to ascertain the coordinates of sources.

2.1 Astrometry

Astrometry is the science of positional astronomy, which measures the location of a celestial object and its movement within the plane of the sky (Endl and Cochran [2007](#)). In terms of this project, astrometry is the process of converting pixel coordinates of our FITS file to a World Coordinate System (WCS), namely the International Celestial Reference System (ICRS). This is done using the website nova.astrometry.net, which takes the input FITS file and gives the output file with full WCS information and orientation based on a method which involves using four reference stars and their relative positions to find a hypothesized alignment with the sky, which is then verified by a Bayesian method outlined in Lang et al. [2010](#).

2.2 Bias and Flat Correction

Bias frames model the sensor's consistent underlying signals, such as the dark current. Flat frames model the defects which alter light transmission through the optical path. Defects such as vignetting and dust are removed during Flat correction. Bias frames are typically short exposures taken with the camera shutter closed. Whereas, flat frames are usually taken by pointing the telescope at a uniformly illuminated surface or the twilight sky.

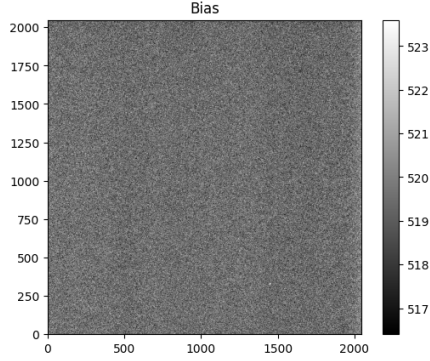


Figure 2.1: Example Bias Frame

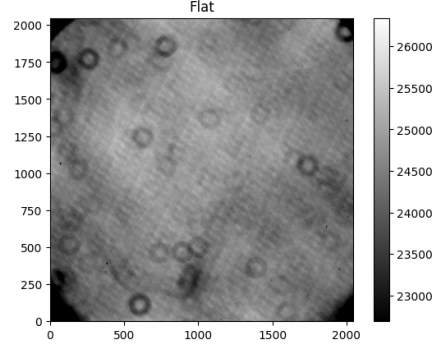


Figure 2.2: Example Flat Frame

In this work, we have also accounted for cosmic rays using the L.A.Cosmic algorithm which is detailed in van Dokkum [2001](#). To clean an image, we use the following -

$$\frac{Image - Bias}{Flat} \quad (2.1)$$

After which, we do the cosmic ray subtraction.

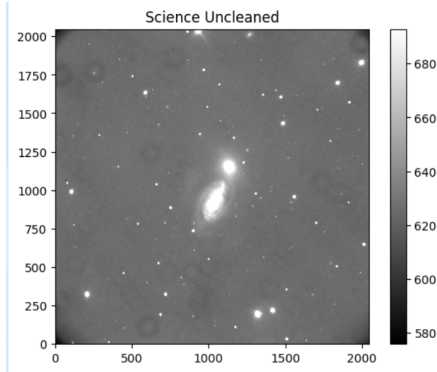


Figure 2.3: Example uncleaned image

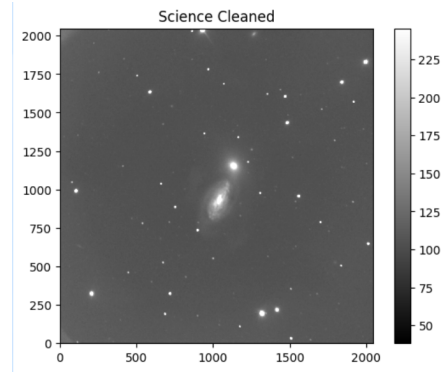


Figure 2.4: Example cleaned image

2.3 Background Subtraction

In this work, we have used the `photutils` python package to do the background subtraction. Specifically, we use the `Background2D` class to estimate a 2D background and background RMS noise in the image. This is done by first defining a certain mesh size in the class, in our case a 50x50 pixel mesh, after which a background estimation method subclass is used. We use the `MedianBackground` class, which finds the median pixel value in the defined mesh.

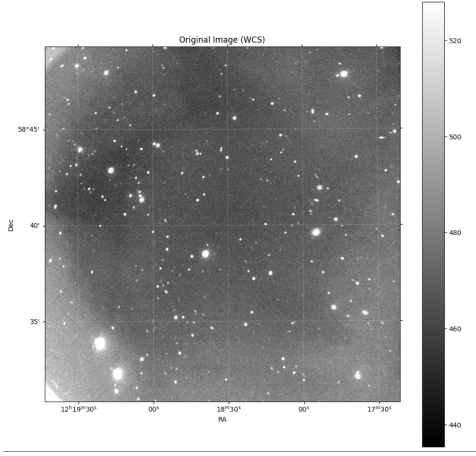


Figure 2.5: Example image without background subtraction

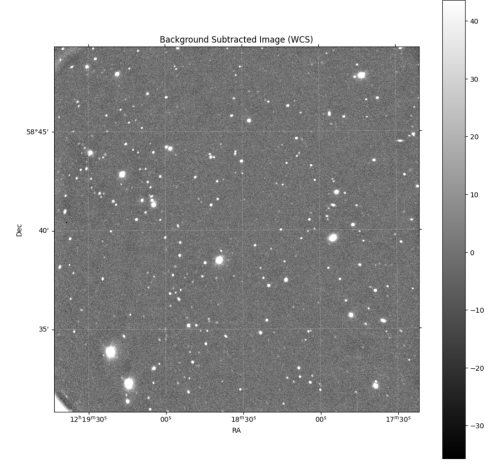


Figure 2.6: Example background subtracted image

Following this, we set a median filter of 3x3 pixels which smoothens out the 2D background across the meshes. Finally, we take the estimated background and subtract it from our data to get background-subtracted data. The same 2D background class also automatically determines the background RMS noise which can then be used in source detection.

2.4 Source Detection

We have done source detection also using the `photutils` python package. The process starts with setting a detection threshold, above which pixels will be considered for processing as source pixels. This is generally set at some multiple of the background RMS noise. We have set the threshold at 3 times the RMS.

Following this, we use a smoothing kernel to help enhance source detection, which is the `make_2dgaussian_kernel` class in our case. This class enables us to smooth over the image at an FWHM of 3 pixels and a kernel size of 5x5 pixels. This kernel is then used with the `convolve` class to perform the convolution of the image, which enhances sources while smoothing over the noise.

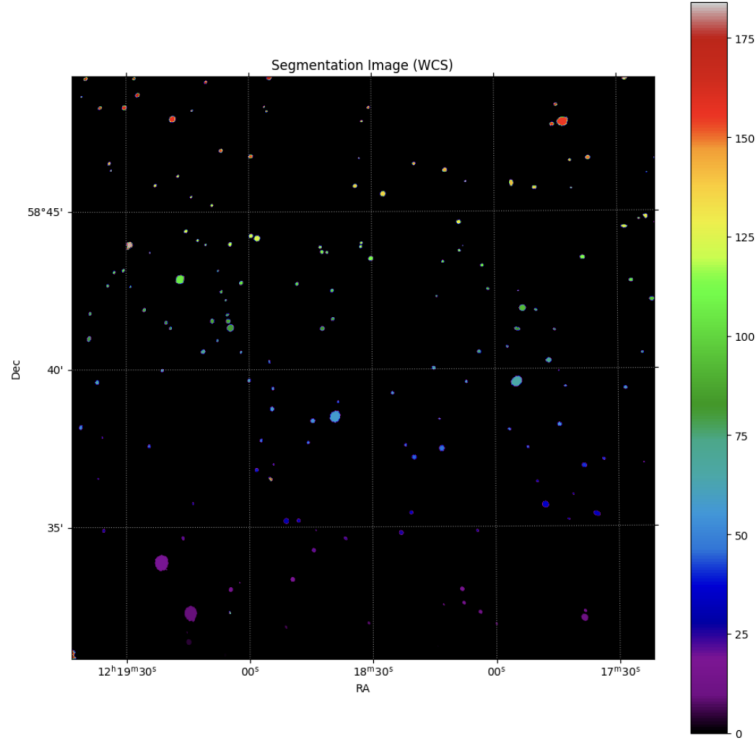


Figure 2.7: Example Deblended Segmentation Image

We then take the convolved data and apply the `detect_sources` class, which picks out groups of connected pixels above the set detection threshold. We set the minimum pixel parameter at 5 pixels for more efficient detection. This class then returns a segmentation image, after which we apply the `deblend_sources` class, which splits blended sources or sources which may be overlapped and labelled as a single source. This is done by analysing segments using multi-thresholding the image, which further segments an image and set various thresholds for each segment, along with a technique called watershed segmentation (Vincent and Soille [1991](#)), which analyses the image as a topographical map and essentially tries to find a saddle point in between overlapping sources.

2.5 Source Catalogue

Creating a source catalogue follows from detecting sources, where we simply assign an index to each of the sources along with the coordinates of the centre of the source found during detection.

Chapter 3

Photometry

This chapter discusses the aperture photometry methods we have used, along with some results found.

3.1 Defining Apertures

To accurately capture the total flux from a given source, it is important to define the aperture for it. We have used a few methods to do this, such as Gaussian fit, Voigt profile fit, and Kron apertures. Both the Gaussian fit and the Voigt fit have similar first steps, which consist of creating consistently larger concentric circular apertures and then finding the surface brightness (flux per unit area) for that aperture radius. The surface brightness is then plotted against the radius of the corresponding aperture, called a radial profile, which is then fitted to find the full width at half-maximum for an appropriate radius.

We then fit with both a Gaussian and a Voigt profile. A Voigt profile is given by a convolution of a Cauchy-Lorentz distribution and a Gaussian distribution.

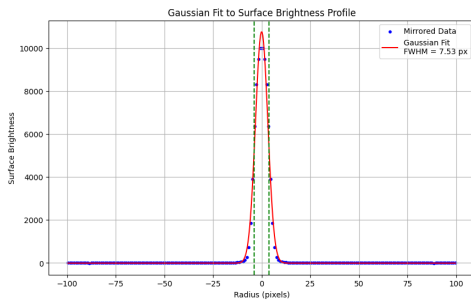


Figure 3.1: Surface Brightness Profile of a source with fitted Gaussian

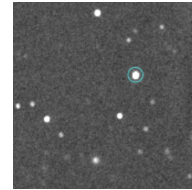


Figure 3.2: Aperture Plotted on the source (Gaussian)

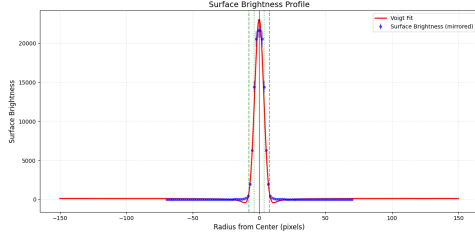


Figure 3.3: Surface Brightness Profile of a source with Voigt fit

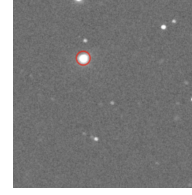


Figure 3.4: Aperture Plotted on the source (Voigt)

On the other hand, the Kron aperture is an adaptive elliptical aperture, which is computed from the centroid and first moment of light distribution, detailed in Kron 1980. It is typically useful for elliptical or extended sources. Kron apertures can be found rather easily using the `SourceCatalog` class in `photutils`. Although the Kron aperture has been proven to capture 90% of flux for a source, scaling the Kron aperture to 2.5 times has been claimed to capture 94% of the flux (Bertin and Arnouts 1996).

3.2 Calibration

Calibration is an essential part of relative photometry. In this work, the method we have used includes using sources of known magnitude to find unknown magnitudes of interest. This involves first finding sources in the field whose magnitude is known to us. We do this using the SIMBAD database (Wenger et al. 2000). Then we plot the flux of these sources against the known magnitude and fit the curve. We can then extrapolate the magnitudes of our unknown sources from this fit.

3.3 Case of Seyfert 2 galaxy with identifier MCG 01-05-47

A galaxy of interest that we studied to find its magnitude was a Seyfert 2 galaxy with the closest identifier being MCG 01-05-47 at a distance of 6.49 arcseconds. In the case of this field, we find that it is quite noisy after bias and flat correction, and even post background subtraction, possibly due to dew.

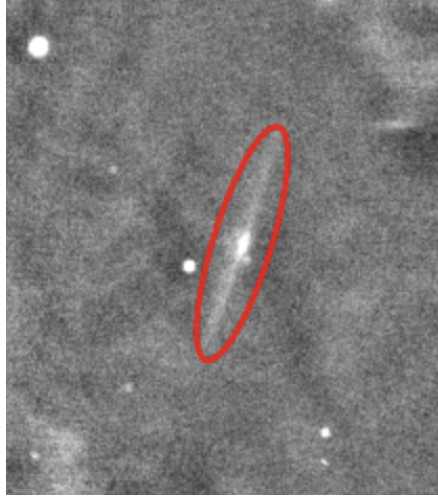


Figure 3.5: Galaxy of interest with Kron aperture

Calibrating this field was also a challenge due to the noise, but since most objects in the field were point-like sources, we could use the Gaussian circular apertures for calibration.

Keeping in mind the large wings of the galaxy, it was important to find the Kron aperture correctly for this source. We scaled the aperture to 2.5 times such that a maximum amount of flux was captured from the wings without capturing much background which would come with increased noise. The flux inherently has an error component of the Poisson noise which is given by -

$$\sigma_N = \sqrt{N} \quad (3.1)$$

where, N is the number of counts and σ_N is the standard deviation of the counts. The magnitude-flux relation and error propagation of the magnitude are given by -

$$m = a \times \log_{10}(flux) + b \quad (3.2)$$

$$\delta m = \sqrt{\left(\frac{\ln 10 \cdot F}{a} \cdot \delta F\right)^2 + (\log_{10}(F) \cdot \delta a)^2 + \delta b^2} \quad (3.3)$$

The calibration for this field was done using 8 known source, which looks like -

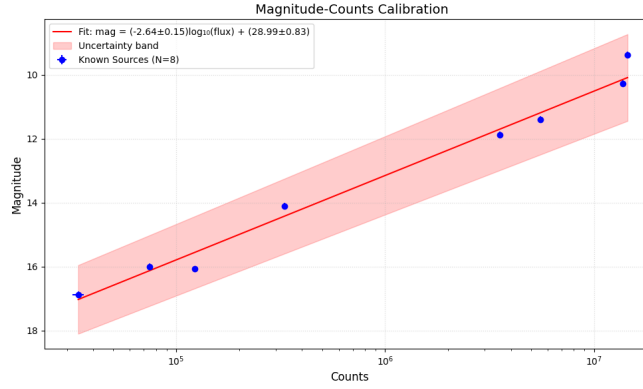


Figure 3.6: Magnitude-Counts calibration for this particular field.

The magnitude found for this particular galaxy using this calibration equation is 14.429 ± 1.172 . The large error is possibly due to the noisy field, which would result in increased Shot noise.

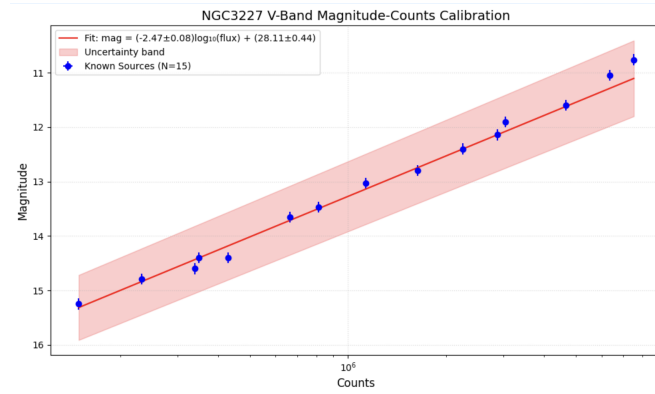


Figure 3.7: Magnitude-Counts calibration for NGC3227 V-Band field.

Another example of a field calibration is shown in [Figure 3.7](#). Obviously, having many more data points such as in this field greatly reduces the width of the uncertainty band.

Chapter 4

Spectral Energy Distributions

A Spectral Energy Distribution is the plot of the flux emitted by a particular object at various wavelengths of light. Usually, an SED would consist of fluxes found for a particular source across the entire electromagnetic spectrum, ranging from the radio to gamma wavelengths. In the context of this study, we do not plot a true SED, but a partial one given the data available to us.

The shape of an SED gives us a lot of critical information about the source being observed. Mainly, we can compare where more of the energy of the emissions lies in the electromagnetic spectrum. For instance, pulsars emit most strongly in the radio range, whereas AGN emit most strongly in X-rays. Peaks of particular sources can tell us about the processes going on in the system being observed.

An SED is analogous to a blackbody radiation curve in the way that it relates brightness to emitted wavelengths. Although the main difference would be the dependence of the blackbody curve on the temperature, whereas an SED would depend on a world of different parameters like object subtypes, dust, other radiation, etc.

In this work, we have data from the U, V, B, and H- α filters. We have plotted SEDs for various objects in different fields and multiple objects in a single field as well.

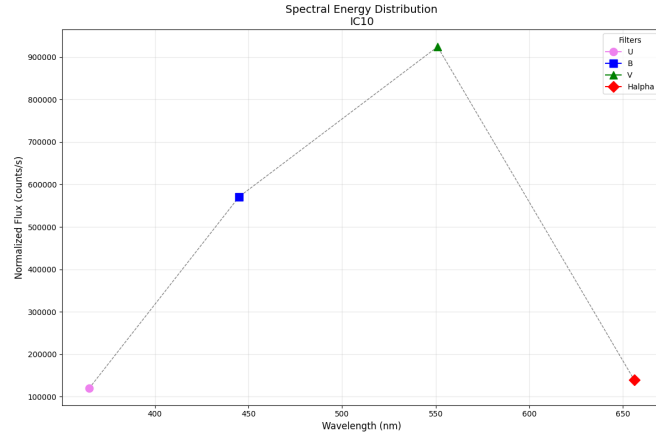


Figure 4.1: Spectral Energy Distribution of IC10.

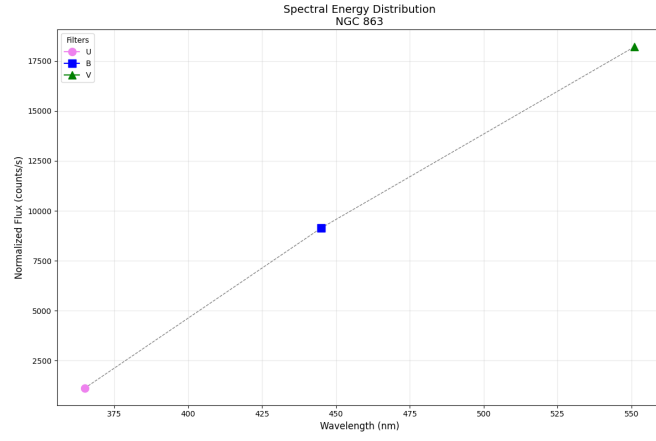


Figure 4.2: Spectral Energy Distribution of NGC 863.

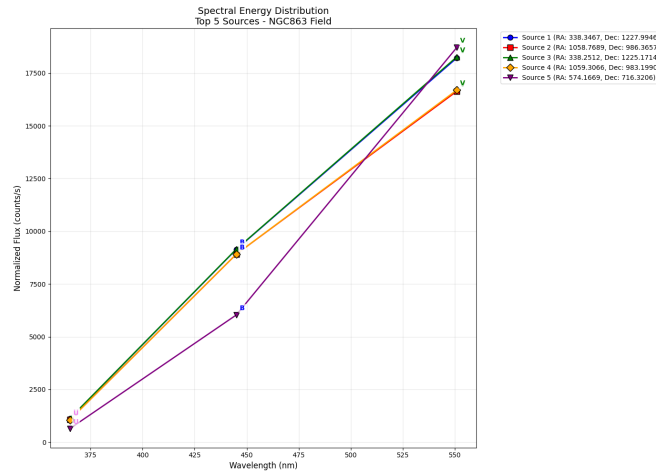


Figure 4.3: Spectral Energy Distribution of Top 5 Sources in the NGC 863 field.

Chapter 5

Star Cluster Analysis of M46

A star cluster is a group of stars held together by self-gravitation. An open cluster is a type of star cluster that is formed from the same molecular cloud. Stars in an open cluster are loosely bound to their cluster by their mutual gravity. They have only been found in spiral or irregular galaxies so far.

5.1 Membership in a Cluster

Membership of a particular star in a cluster can be found in many ways, including using relative distances, age, brightness, and rotation. One of the simplest ways, however, is using proper motion data. Proper motion is the angular speed of an object in the sky. Proper motion is also resolved into RA and Dec components.

On plotting PM-RA (Proper Motion RA) and PM-Dec, it is easy to separate a cluster from background stars, as stars in a cluster will bunch together. This is due to their gravitational binding, which results in them having very similar proper motions.

In the below plot, towards the left, a high density of points is visible, which is the grouping that represents the cluster of interest.

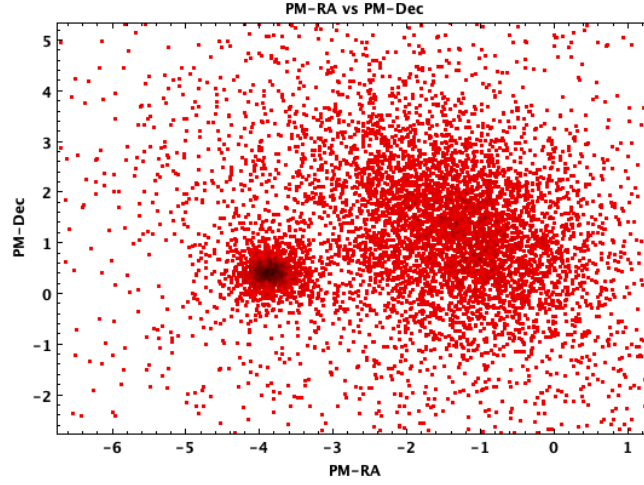


Figure 5.1: Proper Motion visualization

In this work, we used a filtering method which involved finding the mean values of PM-RA and PM-Dec for the cluster (essentially finding where the cluster is centred) and then iteratively filtering sources based on constantly updated statistics and a parallax filter based on the same. Most of the analysis we did for clusters was done using TOPCAT (Taylor 2005) and Python.

Like the Proper Motion plot, a histogram of parallax data can give us a lot of information on membership as well. In the case of this dataset, we see that there is not one distinct peak, which is characteristic of such a cluster.

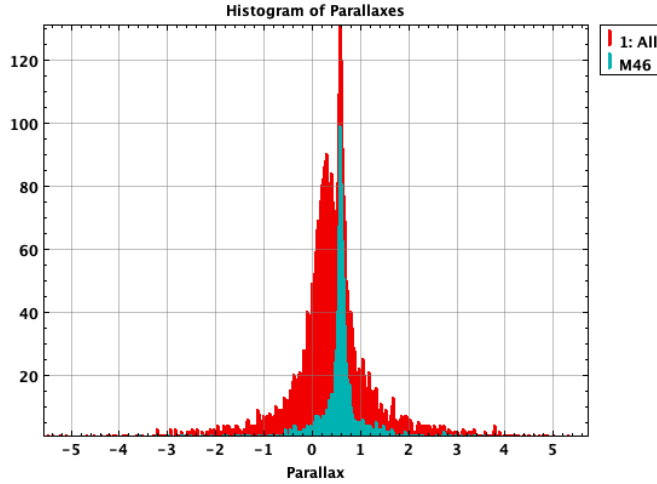


Figure 5.2: Distribution of Parallaxes

5.2 Stellar Evolution

One of the simplest ways of mapping stellar evolution is the Hertzsprung-Russell Diagram, also called a colour-magnitude diagram, depending on the type. In the case of this study, we have plotted the mean G-band magnitude against the difference of the B-band and R-band magnitudes, which is the effective temperature.

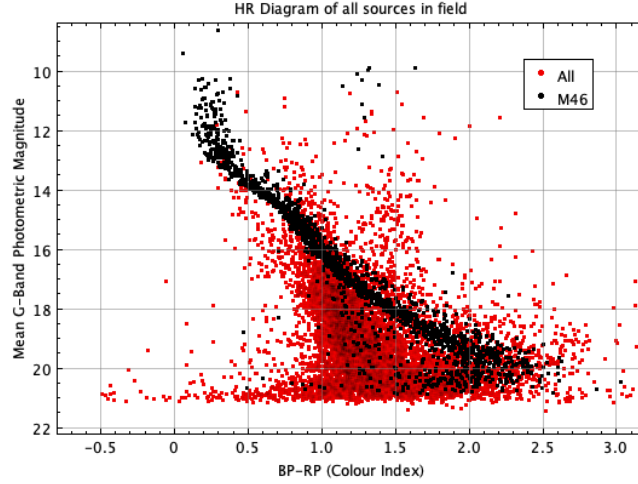


Figure 5.3: HR diagram of all sources

Most stars occupy the central region of the diagram and are called main-sequence stars, which are stars that are fusing hydrogen in their cores. Towards the bottom left are stars that have exhausted or nearly exhausted their fuel, on their way to evolving into white dwarfs. Towards the top of the diagram are stars that are very large, on their way or at the giants/supergiants stage. The hottest burning stars are towards the left side of the diagram, whereas the ones on the right burn at a lower temperature.

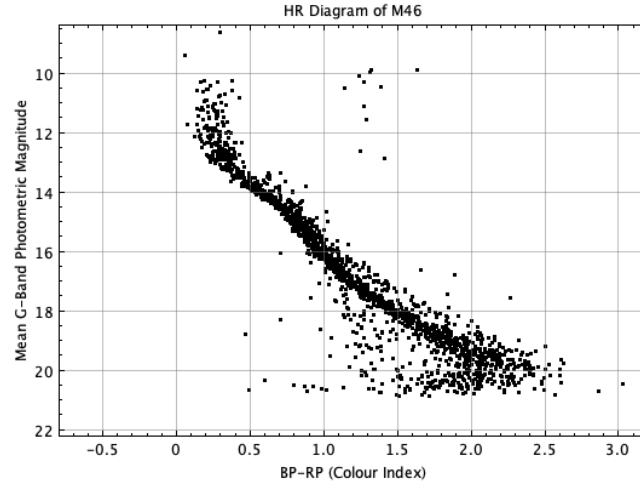


Figure 5.4: HR diagram of cluster members

The above HR diagram is has most members occupying the main sequence region, which is expected as they belong to the same cluster. This is due to the similar ages of the stars, which is a property of cluster members.

Chapter 6

Conclusion

This report details the development and application of techniques for astronomical image processing and data analysis, which provided insights into galactic and stellar properties.

Firstly, a comprehensive data reduction pipeline was developed, proving effective in transforming raw astronomical images into scientifically usable data. This pipeline incorporated essential steps, including astrometry for precise coordinate determination, bias and flat-field corrections to mitigate instrumental artefacts, and advanced background subtraction techniques to isolate celestial sources. Furthermore, the implementation of sophisticated source detection and cataloguing procedures ensured the accurate identification and preliminary characterisation of objects within the processed fields.

Secondly, photometric analysis was rigorously performed using various aperture methods, including Gaussian, Voigt, and adaptive Kron apertures, to accurately quantify the flux and brightness of observed sources. The application of relative photometry, calibrated against known stellar magnitudes from the SIMBAD astronomical database, enabled the precise determination of unknown magnitudes. A notable application involved the Seyfert 2 galaxy with identifier MCG 01-05-47, for which an apparent magnitude of 14.429 ± 1.172 was derived. The associated uncertainty highlighted the challenges posed by noisy observational conditions, underscoring the importance of robust error propagation.

Thirdly, the generation of Spectral Energy Distributions (SEDs) across U, V, B, and H- α photometric bands provided insight into the energy emission characteristics of selected celestial objects. These partial SEDs demonstrated the distribution of emitted energy across different wavelengths, contributing to a preliminary understanding of the

physical processes governing these sources.

Finally, a significant achievement of this project was the detailed analysis of the M46 open star cluster. A highly effective iterative filtering method was successfully implemented, leveraging the distinct proper motion (PM-RA and PM-DEC) and parallax values characteristic of gravitationally bound cluster members. This method accurately distinguished genuine cluster stars from contaminating field stars by progressively refining the selection based on converging kinematic and spatial properties. The resulting Hertzsprung-Russell (HR) diagram for the identified M46 members exhibited a well-defined main sequence, consistent with a co-eval stellar population, thereby validating the efficacy of the membership selection process and confirming the cluster's fundamental properties.

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